# Univariate Linear Regression

## General Principles

To study relationships between a continuous independent variable and a continuous dependent variable (e.g., height and weight), we can use linear regression. Essentially, we draw a line that passes through the point cloud of the two variables being tested. For this, we need to have:

1) An intercept $\alpha$, which represents the origin of the line,i.e., the expected value of the dependent variable (height) when the independent variable (weight) is equal to zero.
2) A coefficient $\beta$, which informs us about the slope of the line. In other words, it tells us how much Y (height) increases for each increment of the independent variable (weight).
3) A standard deviation term $\sigma$, which informs us about the spread of points around the line, i.e., the variance around the prediction.

## Considerations

> **Note**
>
> - Bayesian models allow us to update our understanding of parameters conditional on an observed data set. This allows us to consider model parameter uncertainty , which quantifies our confidence or uncertainty in the parameters in the form of a posterior distribution . Therefore, we need to declare prior distributions for each model parameter, in this case for: $\alpha$, $\beta$, and $\sigma$.
> - Prior distributions are built following these considerations:
>   - As the data are normalized (see introduction), we can use a Normal distribution for $\alpha$ and $\beta$, with a mean of 0 and a standard deviation of 1. This tends to be a weakly regularizing prior, and weaker priors like a $Normal(0, 10)$ are also possible.
>   - Since $\sigma$ must be strictly positive, we must use a distribution with support on the positive reals, such as the *Exponential* or *Folded-Normal* distribution.

- Gaussian regression deals directly with continuous outcomes, estimating a linear relationship between predictors and the outcome variable without depending on a non linear link function (see introduction). This simplifies interpretation, as coefficients represent direct changes in the outcome variable.

### Example

Below is an example code snippet demonstrating *Bayesian linear regression* using the BayesForge (**BF**) package. Data consist of two continuous variables (height and weight), and the goal is to estimate the effect of weight on height. This example is based on McElreath (2018).

### Python

```python
from BayesForge import bf

# Setup device-------------------------------------------------
m = bf(platform="cpu")

# Import Data & Data Manipulation ------------------------------------------------
# Import
data_path = m.load.howell1(only_path=True)
m.data(data_path, sep=";")
m.df = m.df[m.df.age > 18]  # Subset data to adults
m.scale(["weight"])  # Normalize


# Define model -------------------------------------------------
def model(weight, height):
    a = m.dist.normal(178, 20, name="a")
    b = m.dist.log_normal(0, 1, name="b")
    s = m.dist.uniform(0, 50, name="s")
    m.dist.normal(a + b * weight, s, obs=height)


# Run mcmc -------------------------------------------------
m.fit(model, progress_bar=False)  # Optimize model parameters through MCMC sampling

# Summary -------------------------------------------------
m.summary()  # Get posterior distributions
```

```
An NVIDIA GPU may be present on this machine, but a CUDA-enabled jaxlib is not installed. Fal

bf v 0.0.60 package loaded
jax.local_device_count 16
```

| | mean | sd | eti89_lb | eti89_ub | ess_bulk | ess_tail | r_hat | mcse_mean | mcse_sd |
|---|---|---|---|---|---|---|---|---|---|
| a | 154.65 | 0.28 | 154.21 | 155.10 | 4964.48 | 2964.49 | 1.0 | 0.0 | 0.0 |
| b | 5.81 | 0.29 | 5.35 | 6.26 | 4267.26 | 3162.32 | 1.0 | 0.0 | 0.0 |
| s | 5.14 | 0.20 | 4.83 | 5.47 | 4355.95 | 3149.11 | 1.0 | 0.0 | 0.0 |

**R**

```r
library(BayesForge)
m <- importBF(platform = "cpu")

# Load csv file
m$data(m$load$howell1(only_path = T), sep = ";")

# Filter data frame
m$df <- m$df[m$df$age > 18, ] # Subset data to adults

# Scale
m$scale(list("weight")) # Normalize

# Convert data to JAX arrays
m$data_to_model(list("weight", "height"))

# Define model ------------------------------------------------
model <- function(height, weight) {
    # Parameter prior distributions
    s <- bf.dist.uniform(0, 50, name = "s")
    a <- bf.dist.normal(178, 20, name = "a")
    b <- bf.dist.log_normal(0, 1, name = "b")

    # Likelihood
    bf.dist.normal(a + b * weight, s, obs = height)
}

# Run MCMC ----------------------------------------------
```

```
m$fit(model) # Optimize model parameters through MCMC sampling

# Summary ----------------------------------------------
m$summary()
```

### Julia

```
using PyBayesForge

# Setup device-----------------------------------------------
m = importBF(platform="cpu")

# Import Data & Data Manipulation ----------------------------------------------
# Import
data_path = m.load.howell1(only_path = true)
m.data(data_path, sep=';')
m.df = m.df[m.df.age > 18] # Subset data to adults
m.scale(["weight"]) # Normalize

# Define model -----------------------------------------------
@BF function model(weight, height)
    # Priors
    a = m.dist.normal(178, 20, name = 'a')
    b = m.dist.log_normal(0, 1, name = 'b')
    s = m.dist.uniform(0, 50, name = 's')
    m.dist.normal(a + b * weight , s, obs = height)
end

# Run mcmc -----------------------------------------------
m.fit(model)  # Optimize model parameters through MCMC sampling

# Summary ----------------------------------------------
m.summary() # Get posterior distributions
```

## Mathematical Details

### *Frequentist Formulation*

The following equation describe the frequentist formulation of linear regression:

$$Y_i = \alpha + \beta X_i + \epsilon_i$$

Where:

- $Y_i$ is the dependent variable for observation $i$.
- $\alpha$ is the intercept term.
- $\beta$ is the regression coefficient.
- $X_i$ is the input variable for observation $i$.
- $\epsilon_i$ is the error term for observation $i$, and the vector of the error terms, $\epsilon$, are assumed to be independent and identically distributed.

### *Bayesian Formulation*

In the Bayesian formulation, we define each parameter with priors . We can express a Bayesian version of this regression model using the following model:

$$Y_i \sim \text{Normal}(\alpha + \beta X_i, \sigma)$$

$$\alpha \sim \text{Normal}(178, 20)$$

$$\beta \sim \text{LogNormal}(0, 1)$$

$$\sigma \sim \text{Uniform}(0, 50)$$

Where:

- $Y_i$ is the dependent variable for observation $i$.
- $\alpha$ and $\beta$ are the intercept and regression coefficient, respectively.
- $X_i$ is the independent variable for observation $i$.
- $\sigma$ is the standard deviation of the Normal distribution, which describes the variance in the relationship between the dependent variable $Y$ and the independent variable $X$.

## Model validation tests

- Comparison with Stan: `1.Continuous variable.py` fits the same model with BayesForge and Stan and compares the posterior distributions (KL divergence).
- Documentation examples: the Python, R and Julia examples of this page are run by the documentation tests (`Python/test_model_pages.py`, `R/test_r_model_pages.py`, `Julia/test_julia_model_pages.py`).

## Notes

> **Note**
>
> We observe a difference between the *Frequentist* and the *Bayesian* formulation regarding the error term. Indeed, in the *Frequentist* formulation, the error term $\epsilon$ represents residual fluctuations around the predicted values. This assumption leads to point estimates for $\alpha$ and $\beta$. In contrast, the *Bayesian* formulation treats $\sigma$ as a parameter with its own prior distribution. This allows us to incorporate our uncertainty about the error term into the model.

## Reference(s)

McElreath, Richard. 2018. *Statistical Rethinking: A Bayesian course with examples in R and Stan*. Chapman; Hall/CRC.